



The University of Azad Jammu and Kashmir Muzaffarabad

Name: Jawahir Ali

Roll No: 2024-SE-34

Course: Computer Architecture and Logic Design

Instructor: Engr. Sidra Rafique

Task: Open Ended Lab

Department: Software Engineering

Task No# 1

Parity Generator and Checker

1. 4-bit Parity Generator Circuit

- Inputs: A, B, C, D (4-bit data word).
- Output: Parity Bit (P).
- **Even parity rule:** The parity bit is chosen such that the total number of 1's (data bits + parity) is even.

Boolean Equation:

$$P = A \oplus B \oplus C \oplus D$$

2. Implementation

- Use XOR gates in PLA or a digital logic simulator like Electronic Work Bench..
- Connect inputs A, B, C, D $\rightarrow$ XOR chain $\rightarrow$ Output P.

3. Testing the Generator

- Try all 16 combinations of A, B, C, D.
- Verify that $(A \oplus B \oplus C \oplus D) + P$ always gives **even number of 1s**.

4. Parity Checker Circuit

- Inputs: A, B, C, D, and received Parity Bit (P).
- Output: Error signal (E).

Boolean Equation:

$$E = A \oplus B \oplus C \oplus D \oplus P$$

- If $E = 0 \rightarrow$ No error.
- If $E = 1 \rightarrow$ Error detected.

5. Implementation and Testing

- Connect inputs to XOR chain.
- Apply different data with parity bit.
- Check if circuit correctly detects error when a bit flips.

4-bit Parity Generator Circuit

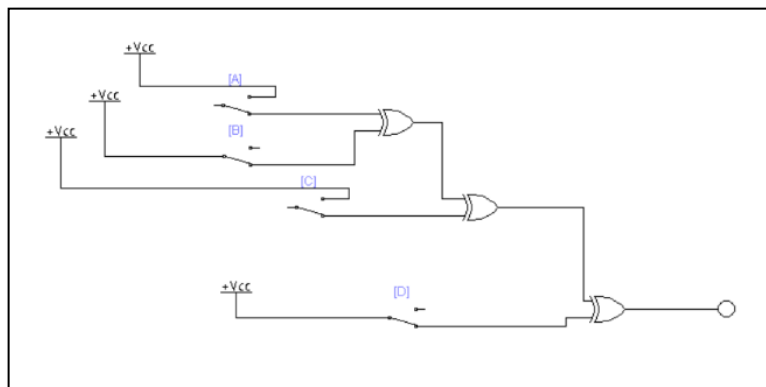


Figure 1: When Switches inputs $(A, B, C, D) = (0, 1, 0, 1)$ then output is 0.

4-bit Parity Generator (Even Parity):

$$\text{Formula: } P = A \oplus B \oplus C \oplus D$$

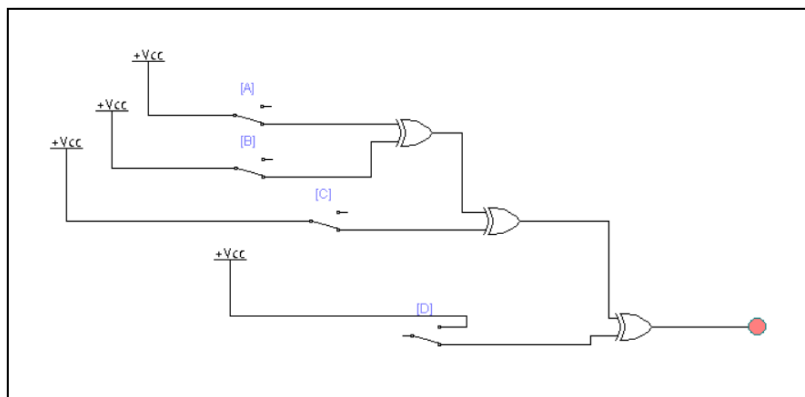
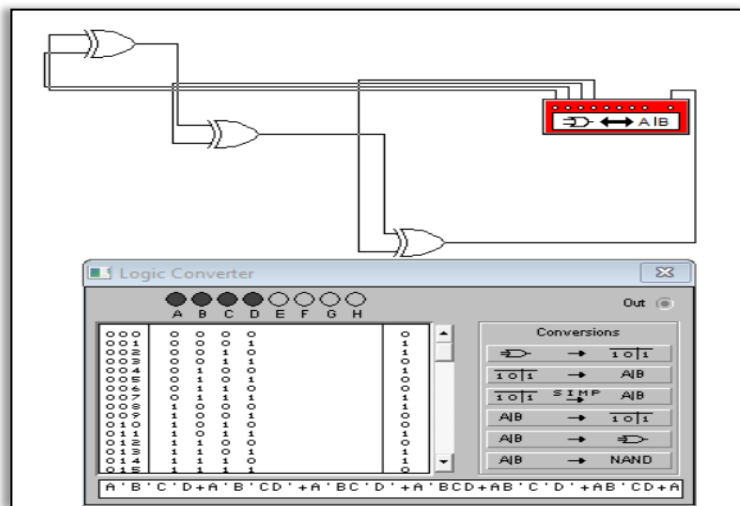


Figure 2 : When three Switches ON and one OFF means all inputs $(A, B, C, D) = (1, 1, 1, 0)$, then output is 1.

Truth Table

A	B	C	D	ParityBit_P	m
0	0	0	0	0	0
0	0	0	1	1	1
0	0	1	0	1	2
0	0	1	1	0	3
0	1	0	0	1	4
0	1	0	1	0	5
0	1	1	0	0	6
0	1	1	1	1	7
1	0	0	0	1	8
1	0	0	1	0	9
1	0	1	0	0	10
1	0	1	1	1	11
1	1	0	0	0	12
1	1	0	1	1	13
1	1	1	0	1	14
1	1	1	1	0	15

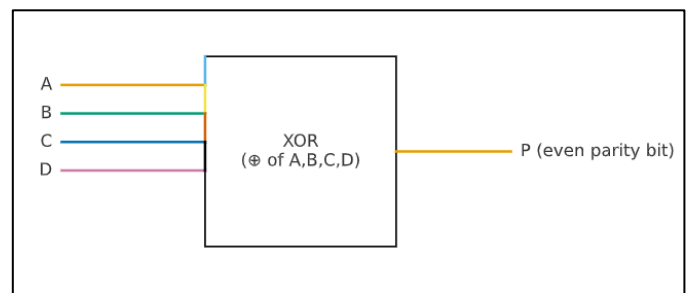
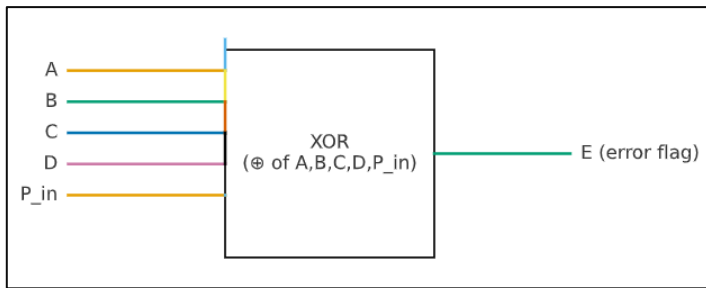


4-Parity Checker (Error Detection)

- Inputs: A, B, C, D, and received Parity Bit (P).
- Output: Error signal (E).

Boolean Equation:

$$E = A \oplus B \oplus C \oplus D \oplus P$$



- If $E = 0 \rightarrow$ No error.
- If $E = 1 \rightarrow$ Error detected.

Implementation and Testing

- Connect inputs to XOR chain.
- Apply different data with parity bit.
- Check if circuit correctly detects error when a bit flips.

4-bit Parity Checker (Error Detection)

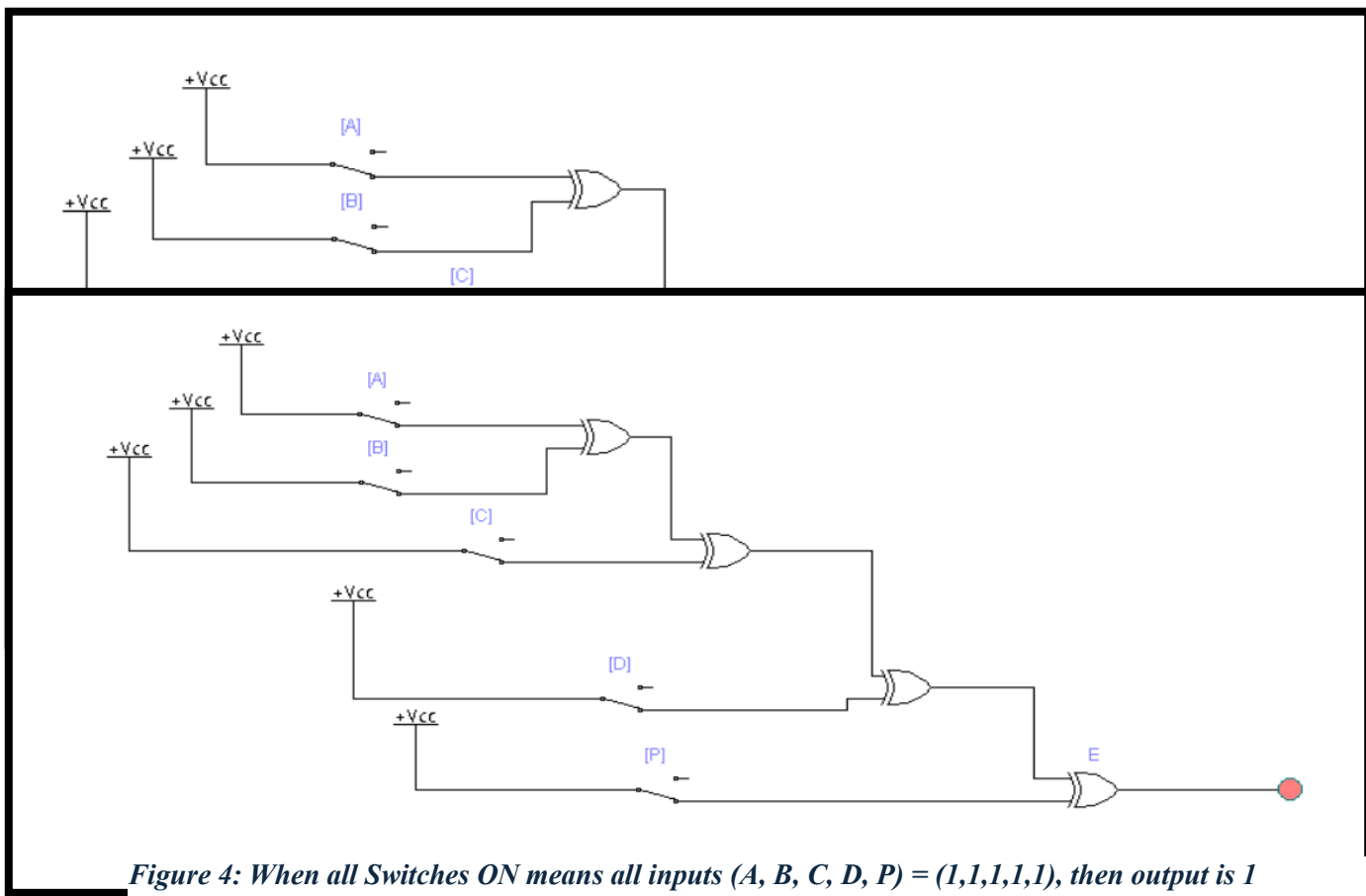


Figure 4: When all Switches ON means all inputs $(A, B, C, D, P) = (1, 1, 1, 1, 1)$, then output is 1

Parity Checker Truth Table

Boolean Equation:

$$E = A \oplus B \oplus C \oplus D \oplus P$$

Inputs: A, B, C, D, P → Output: E (Error)

A	B	C	D	P (Expected)	E (Output)	Meaning
0	0	0	0	0	0	No error
0	0	0	0	1	1	Error
1	0	1	0	0	0	No error
1	0	1	0	1	1	Error
1	1	1	1	0	0	No error
1	1	1	1	1	1	Error

Task 2

Low-Level Program for Shift Instructions

- SHL (Shift Left Logical) – Multiplies number by 2.
- SHR (Shift Right Logical) – Divides unsigned number by 2.
- SAR (Shift Arithmetic Right) – Divides signed number by 2, keeps sign.
- SAL (Shift Arithmetic Left) – Similar to SHL.

MOV AL, 00001101b ; Load data = 13 (binary)

SHL AL, 1 ; Shift Left → result = 26

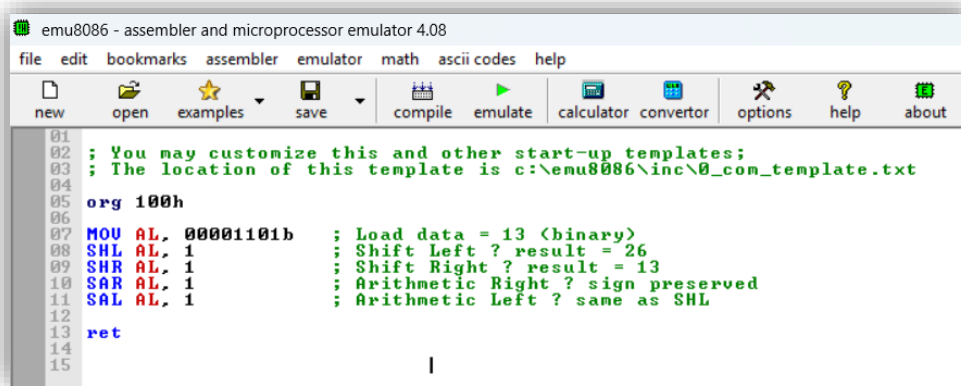
SHR AL, 1 ; Shift Right → result = 13

SAR AL, 1 ; Arithmetic Right → sign preserved

SAL AL, 1 ; Arithmetic Left → same as SHL

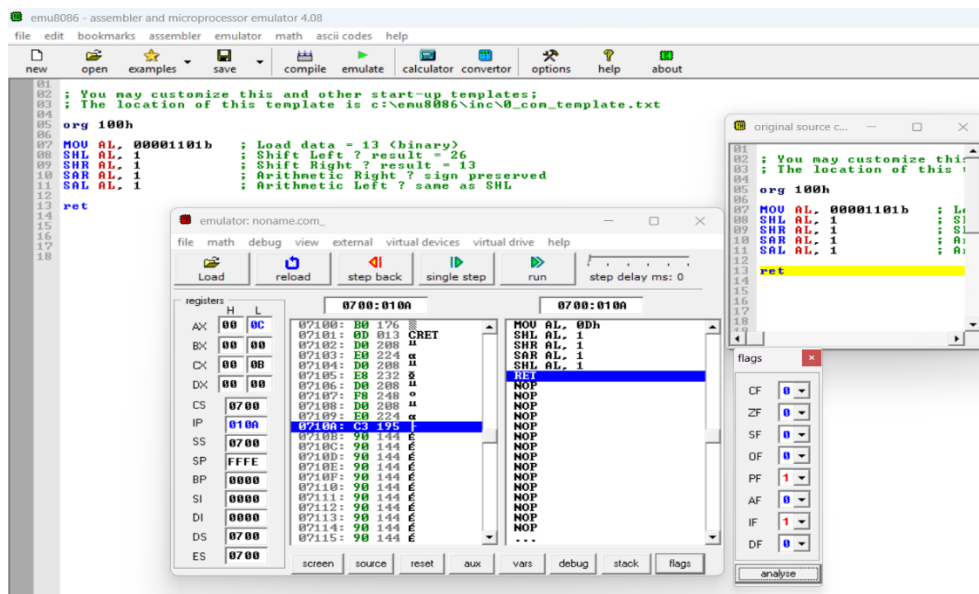
By using Emulator

Step #01:



```
01 ; You may customize this and other start-up templates;  
02 ; The location of this template is c:\emu8086\inc\0_com_template.txt  
03  
04  
05 org 100h  
06  
07 MOV AL, 00001101b ; Load data = 13 (binary)  
08 SHL AL, 1 ; Shift Left ? result = 26  
09 SHR AL, 1 ; Shift Right ? result = 13  
10 SAR AL, 1 ; Arithmetic Right ? sign preserved  
11 SAL AL, 1 ; Arithmetic Left ? same as SHL  
12  
13 ret  
14  
15
```

Result:



Deliverables

- Working Circuits
 - 4-bit parity generator and checker implemented in simulator.

• Truth Table & Boolean Equations

A	B	C	D	Parity P = $A \oplus B \oplus C \oplus D$
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0

Boolean Equation:

$E = A \oplus B \oplus C \oplus D \oplus P$